

DevSpark Round 2 — Challenge

Problem Statements & AI Prompt Engineering Framework

Event	DevSpark — Inter-College Innovation & AI Sprint
Organized By	Team Stack Tracers Department of Computer Science & Engineering
Round Format	Round 2 (DevSprint: Rapid AI-Augmented Solution Build)

1. AI Tooling Framework & Integrity Rules

Generative AI tools (ChatGPT, Gemini, Claude, GitHub Copilot, v0, Cursor and all other AI tools) are explicitly approved for scaffolding, boilerplate generation, and rapid UI/logic synthesis. Rote memorization is not assessed; instead, evaluation focuses on architectural clarity, functional stability, prompt orchestration, and technical defense.

Core Policy	Operational Requirement
The "Zero-Black-Box" Rule	Teams must independently understand and explain every function, data structure, state mutation, and API call during jury examination. Unfamiliarity with AI-generated components results in severe defense deductions.
Mandatory Prompt Logging	Every submission must include an AI_PROMPT_LOG.md file documenting the top 3–5 foundational system, scaffolding, or refinement prompts used during development.
Live Dynamic Execution	Static wireframes, hardcoded non-reactive templates, or purely visual mockups will receive 0 points for functionality. Prototypes must accept user inputs and yield real-time programmatic outputs.

2. The 5 Rapid-Build Problem Statements

Each team must choose **exactly one** track at the start of the 90-minute build window:

Track & Title	Context & Challenge	Mandatory Core Deliverables
Track 1: Campus Flow & Resource Optimization Real-Time Dynamic Resource Conflict & Slot Optimizer	Campuses face frequent booking clashes over shared facilities (seminar halls, computing labs, A/V gear, sports facilities).	1. Interactive venue/slot reservation portal. 2. Automated conflict detection engine for schedule overlaps. 3. Intelligent fallback suggester recommending nearest open slots/venues. 4. Admin dashboard with real-time approval/cancellation controls.
Track 2: Academic & Exam Intelligence Dynamic Syllabus Parser & Bloom's Taxonomy Quiz Engine	Educators and students spend substantial effort converting extensive lecture materials into balanced, multi-tier practice assessments.	1. Document intake parser for syllabus notes, markdown, or text. 2. LLM prompt workflow generating 5–10 questions tiered by Bloom's Taxonomy (Recall, Application, Code Tracing). 3. Interactive quiz UI with live countdown, instant submission, and scoring. 4. Downloadable conceptual weakness/strength summary.
Track 3: Multidisciplinary Peer Collaboration SkillBarter — Reciprocal Campus Teammate Matchmaker	Interdisciplinary hackathons require paired technical, creative, and managerial skills, but discovering campus collaborators remains siloed.	1. Student profile registry specifying offered and sought competencies. 2. Bilateral compatibility algorithm calculating reciprocal match scores. 3. Instant team formation trigger creating shared collaborative cards & milestone lists. 4. Categorical filtering across tech, design, and product management

Track & Title	Context & Challenge	Mandatory Core Deliverables
		tags.
Track 4: Student Finance & Smart Campus Living CampusSplit & Smart Canteen Expense Balancer	Shared expenditures across group food orders, project hardware costs, and hostel utilities create tangled debt cycles.	- Expense-logging engine handling itemized, uneven, and recurring splits. - Graph-based debt minimization algorithm minimizing total settlement transactions across N parties. - Dynamic UPI payment link / QR-code settlement generator. - Visual expenditure distribution analytics dashboard.
Track 5: Campus Safety & Emergency Incident Tracker SafeCampus — Geo-Tagged Anonymous Alert & Incident Dispatcher	Prompt reporting of campus infrastructure hazards, medical contingencies, and safety issues requires instantaneous triage and routing.	10-second rapid incident dispatch form with optional anonymous mode. - Simulated geo-location tagging (e.g., Block A, Lab 3) with urgency priority indicators. - Real-time admin emergency triage desk with dynamic lifecycle statuses ("Dispatched", "Under Review", "Resolved"). - Simulated emergency SOS contact broadcast system.

3. Technical Deliverables & Submission Requirements

Deliverable File / Item	Specification & Contents
1. Working Source Code	Complete source repository (Web application, Mobile client, or CLI tool) containing all components required for execution.

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2. README.md	Clear, step-by-step instructions to install dependencies and run the project locally, or a working hosted link (Vercel, Streamlit, Netlify).
3. AI_PROMPT_LOG.md	A transparent log containing 3–5 primary system, architecture, or logic prompts used across the build to verify prompt engineering proficiency.